# FRC SENSOR CAN — REVISION D REGULATED PROTOTYPE

Three fixed-threshold digital inputs. 6–22 V input; regulated 12 V and 5 V sensor power.
STM32F042 + TCAN3413. Digital status and edge counts over a custom team CAN protocol.
Not a CTRE Phoenix-compatible device. Inspect and bench-test prototypes before robot use.

Battery entry and internal supplies

File: power.kicad_sch

Regulated 12 V synchronous buck-boost

File: regulated_12v.kicad_sch

Protected sensor power rails

File: sensor_power.kicad_sch

Controller, clock, debug and status

File: controller.kicad_sch

CAN physical interface and termination

File: can.kicad_sch

Input 1: fixed-threshold digital input

File: input1.kicad_sch

Input 2: fixed-threshold digital input

File: input2.kicad_sch

Input 3: fixed-threshold digital input

File: input3.kicad_sch

Original design — custom WPILib CAN protocol

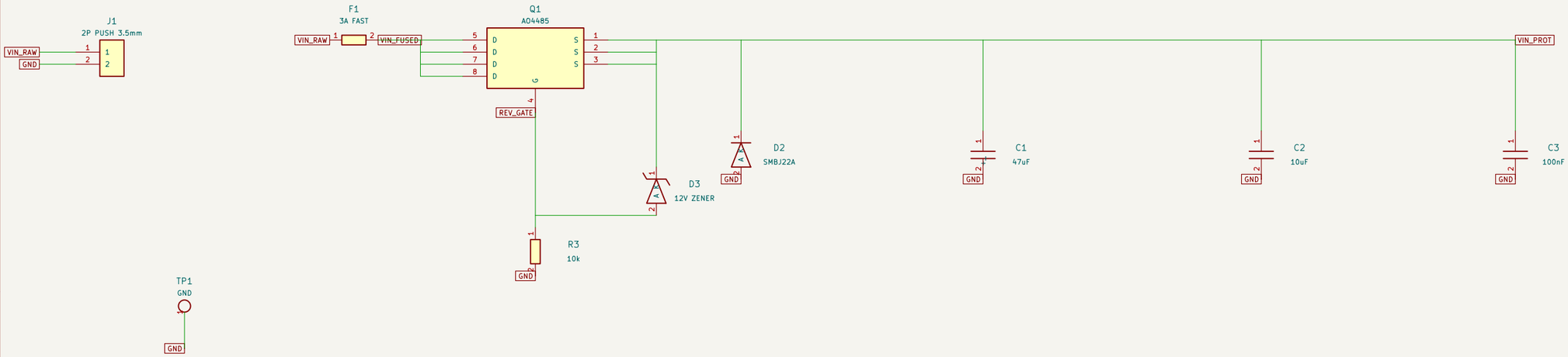
Sheet: /
File: FRC_Sensor_CAN.kicad_sch

**Title: FRC Sensor CAN**

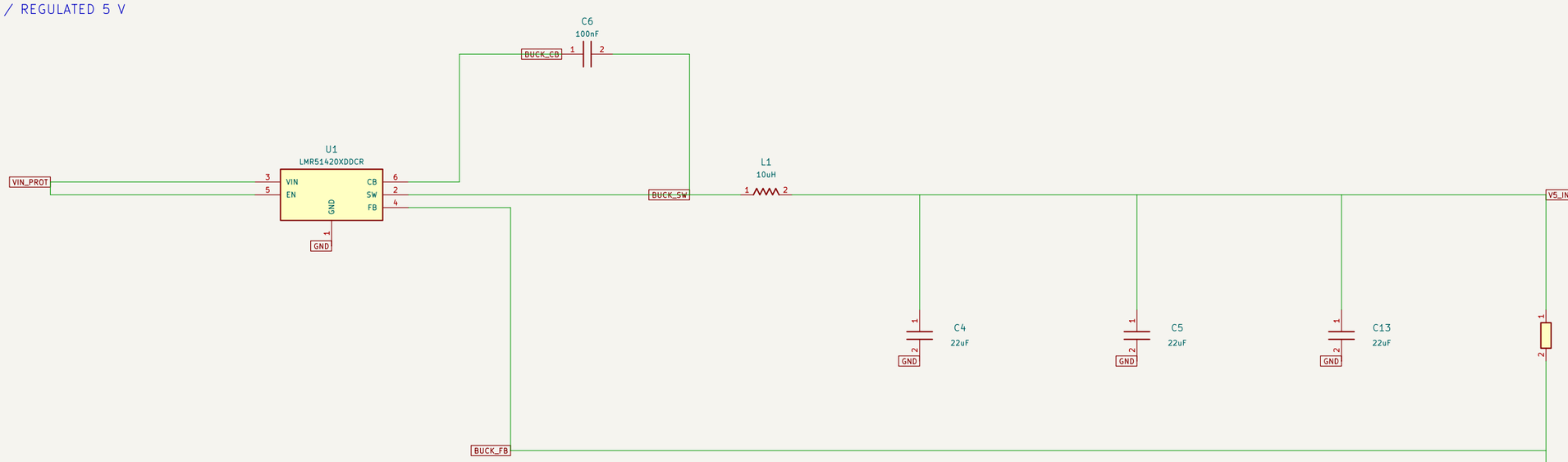
| Size: A3 | Date: 2026-09-08 | **Rev: D** |
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| KiCad E.D.A. 10.0.6 | | Id: 1/9 |

# Battery entry and internal supplies

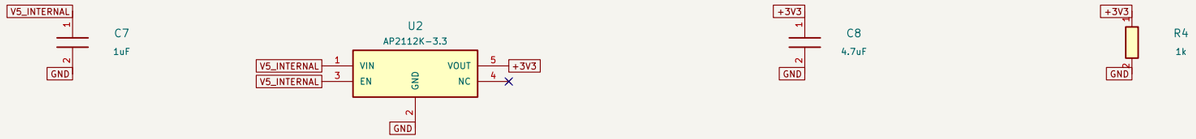
## PROTECTED BATTERY ENTRY




## 500 kHz BUCK / REGULATED 5 V




## 3.3 V LOGIC SUPPLY




6–22 V input target. Single buck: 5.07 V nominal.
12 V buck-boost supply is on the next sheet.
R1/R2 = 102k / 13.7k, 0.1%. Buck-boost uses external bias from this 5 V rail.
TP1 is ground; use selectors/capacitor pads for rail checks.

Prototype: bench validation required before robot use

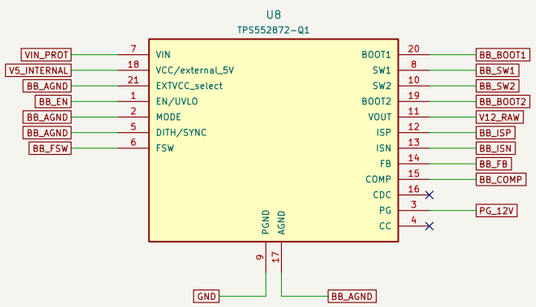
**Original sensor interface design**

Sheet: /Battery entry and internal supplies/
File: power.kicad_sch

***Title: SC12 – Battery entry and internal supplies***

| Size: A3 | Date: 2026-09-08 | **Rev: D** |
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# Regulated 12 V synchronous buck–boost























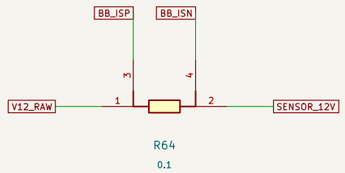








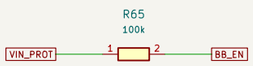






























12.011 V nominal: 300 mA useful shared sensor load. R64 sets approximately 0.5 A output current limit.
MODE and DITH grounded: automatic PFM for light–load efficiency. 24.9k sets 781 kHz nominal.
VCC supplied by the 5 V buck: EXTVCC is a LOGIC SELECT input tied low. No extra regulator.
Join AGND to PGND only at C68 negative pad using NT1. Keep ISP/ISN Kelvin traces separate from load copper.
PG goes to MCU PA3 with internal pull–up. Power–good is an indication, not a calibrated rail measurement.
Verify startup, current limiting, compensation, transients and temperature on assembled prototypes.

Prototype: bench validation required before robot use

**Original sensor interface design**

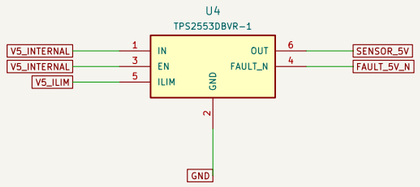
Sheet: /Regulated 12 V synchronous buck–boost/
File: regulated_12v.kicad_sch

***Title: SC12 – Regulated 12 V synchronous buck–boost***

| Size: A3 | Date: 2026–09–08 | **Rev: D** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/9 |

# Protected sensor power rails

HARDWARE CURRENT LIMITING; NO FIRMWARE ENABLE













5 V sensor output: TPS2553-1 latch-off version, EN tied to input.
59k current-limit resistor. Cycle board power to clear overload.
FAULT_N goes to MCU PA4, with a firmware-enabled pull-up.
Design target: 300mA combined load on this rail, not per connector.
Do not back-power either sensor supply output.

Prototype: bench validation required before robot use
**Original sensor interface design**

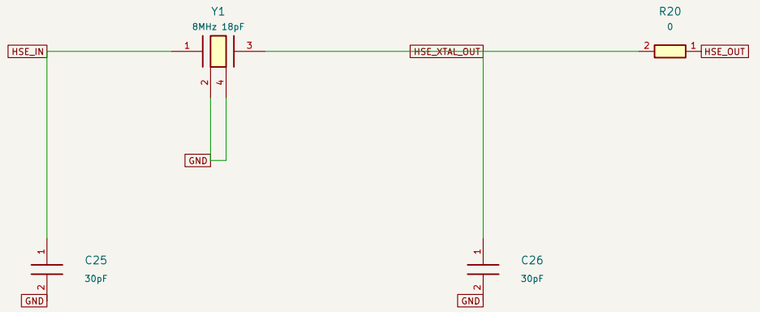
Sheet: /Protected sensor power rails/
File: sensor_power.kicad_sch

***Title: SC12 – Protected sensor power rails***

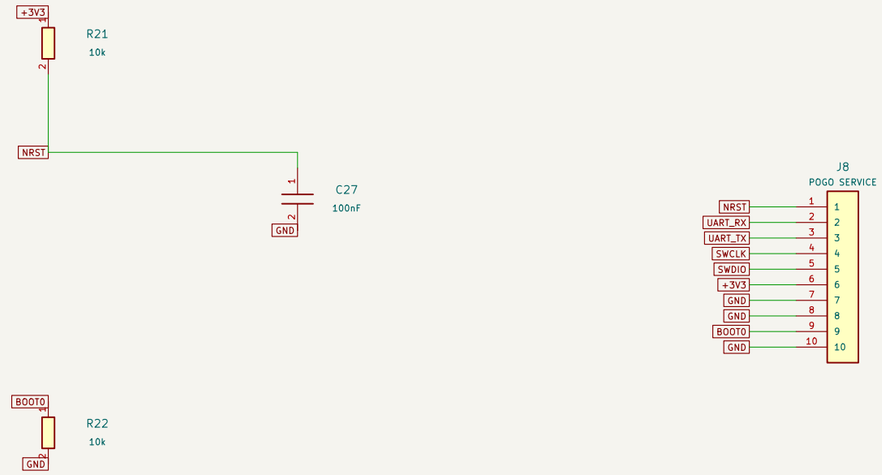
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| Size: A3 | Date: 2026-09-08 | **Rev: D** |
| KiCad E.D.A. 10.0.6 | | Id: 4/9 |

# Controller, clock, debug and status

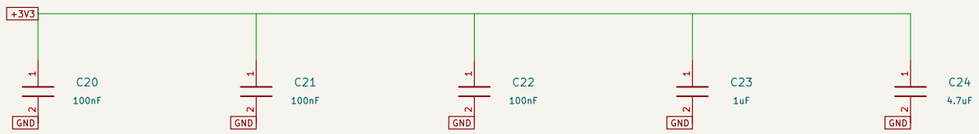
## 8 MHz CRYSTAL / PLL x6

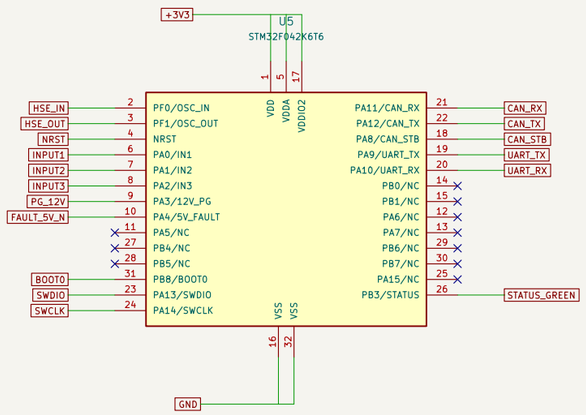



## RESET / BOOT / SERVICE




## LOCAL DECOUPLING




TC2050-IDC-NL mechanical footprint; CUSTOM electrical pinout.
Pin 6 is 3.3 V reference OUTPUT, never a programming power input.
Unused GPIOs are unconnected and initialized to analog mode.
PA3 monitors 12 V PG; PA4 monitors active-low 5 V fault. No rail ADCs.

Prototype: bench validation required before robot use

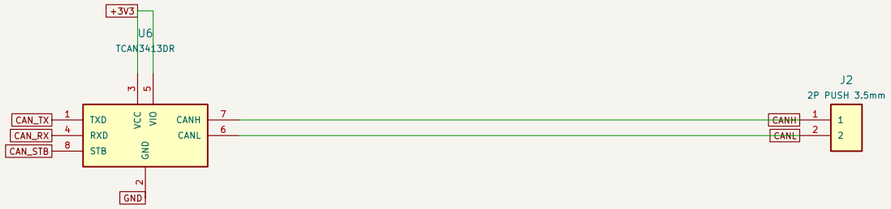
**Original sensor interface design**

Sheet: /Controller, clock, debug and status/
File: controller.kicad_sch

***Title: SC12 – Controller, clock, debug and status***

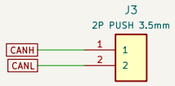
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| Size: A3 | Date: 2026-09-08 | **Rev: D** |
| KiCad E.D.A. 10.0.6 | | Id: 5/9 |

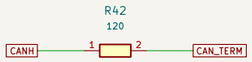
# CAN physical interface and termination

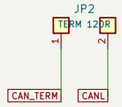


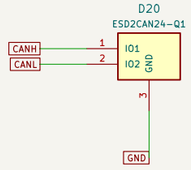


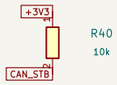


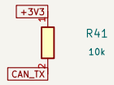


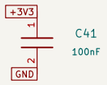



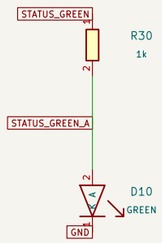







1 Mbit/s. Bridge JP2 only at a physical CAN trunk end.
TCAN3413 uses 3.3 V for both VCC and VIO.
STB pull-up keeps the bus silent during MCU reset.
CAN ground is shared through the power connection.

## SINGLE STATUS INDICATOR




Prototype: bench validation required before robot use
**Original sensor interface design**

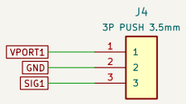
Sheet: /CAN physical interface and termination/
File: can.kicad_sch

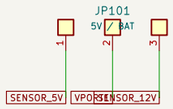
***Title: SC12 – CAN physical interface and termination***

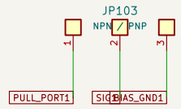
| Size: A3 | Date: 2026-09-08 | **Rev: D** |
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| KiCad E.D.A. 10.0.6 | | Id: 6/9 |

# Input 1: fixed-threshold digital input

SOLDER CONFIGURATION: SELECT POWER AND SENSOR OUTPUT TYPE





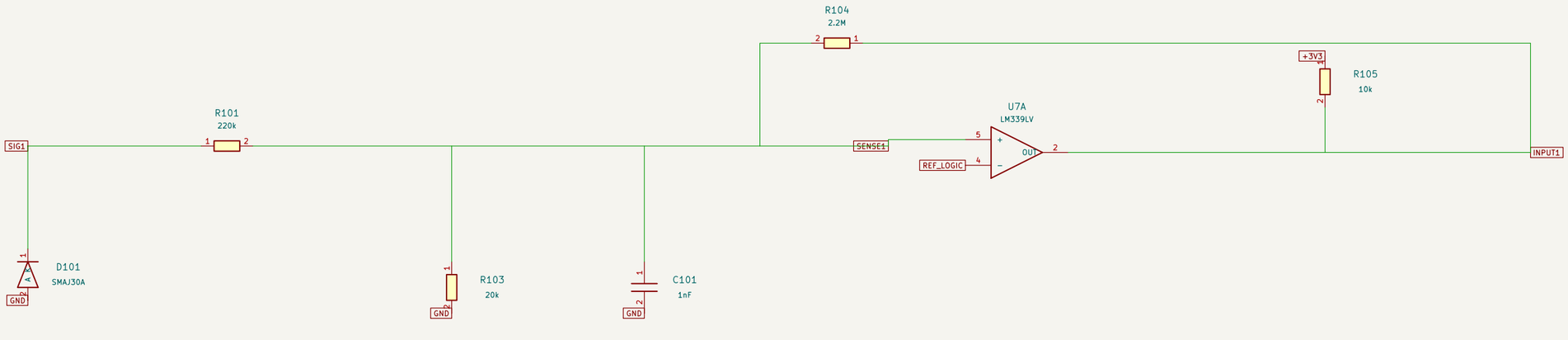








POWER: bridge 1-2 for 5 V OR 2-3 for regulated 12 V. NEVER bridge all three.
BIAS: bridge 1-2 for NPN, 2-3 for PNP, leave open for push-pull/external bias.
All selectors are open as fabricated. Voltage-tolerant NPN output required for selected pullup.



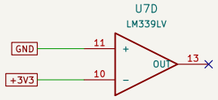


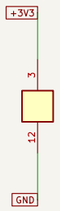











U7E
LM339LV




0-30 V signal design range; software debounce follows the hardware filter.
Nominal thresholds near 2.0 V rising / 1.65 V falling. Final bounds are in electrical_checks.json.
No analog measurement, 1.8 V logic guarantee or CTRE protocol compatibility.

Prototype: bench validation required before robot use
**Original sensor interface design**

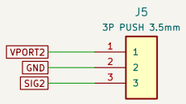
Sheet: /Input 1: fixed-threshold digital input/
File: input1.kicad_sch

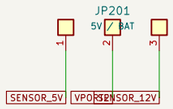
***Title: SC12 – Input 1: fixed-threshold digital input***

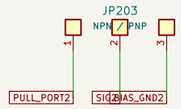
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| KiCad E.D.A. 10.0.6 | | Id: 7/9 |

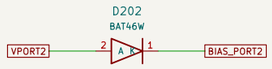
# Input 2: fixed-threshold digital input

SOLDER CONFIGURATION: SELECT POWER AND SENSOR OUTPUT TYPE



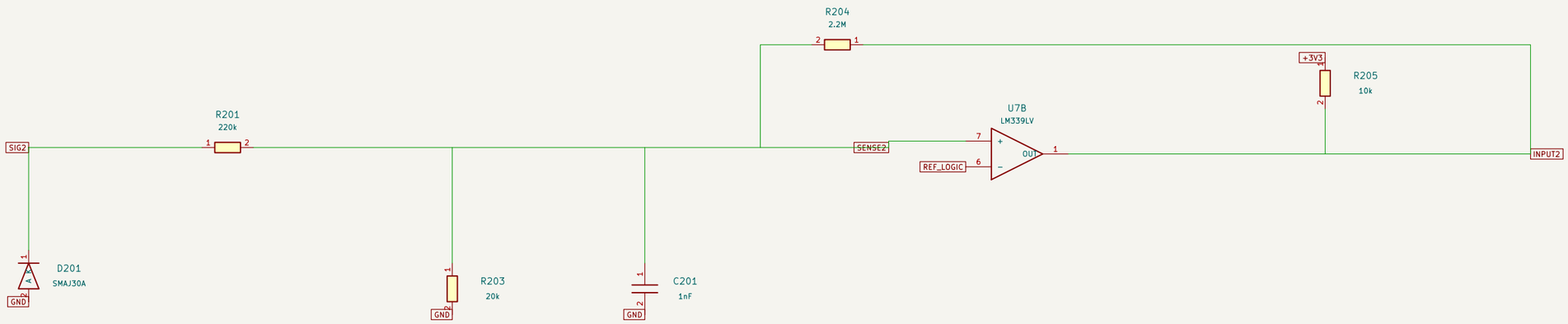










POWER: bridge 1–2 for 5 V OR 2–3 for regulated 12 V. NEVER bridge all three.
BIAS: bridge 1–2 for NPN, 2–3 for PNP, leave open for push-pull/external bias.
All selectors are open as fabricated. Voltage-tolerant NPN output required for selected pullup.




The reference divider/filter and comparator power connections are on input1.
One fixed logic-compatible threshold for all three channels; no industrial threshold selector.

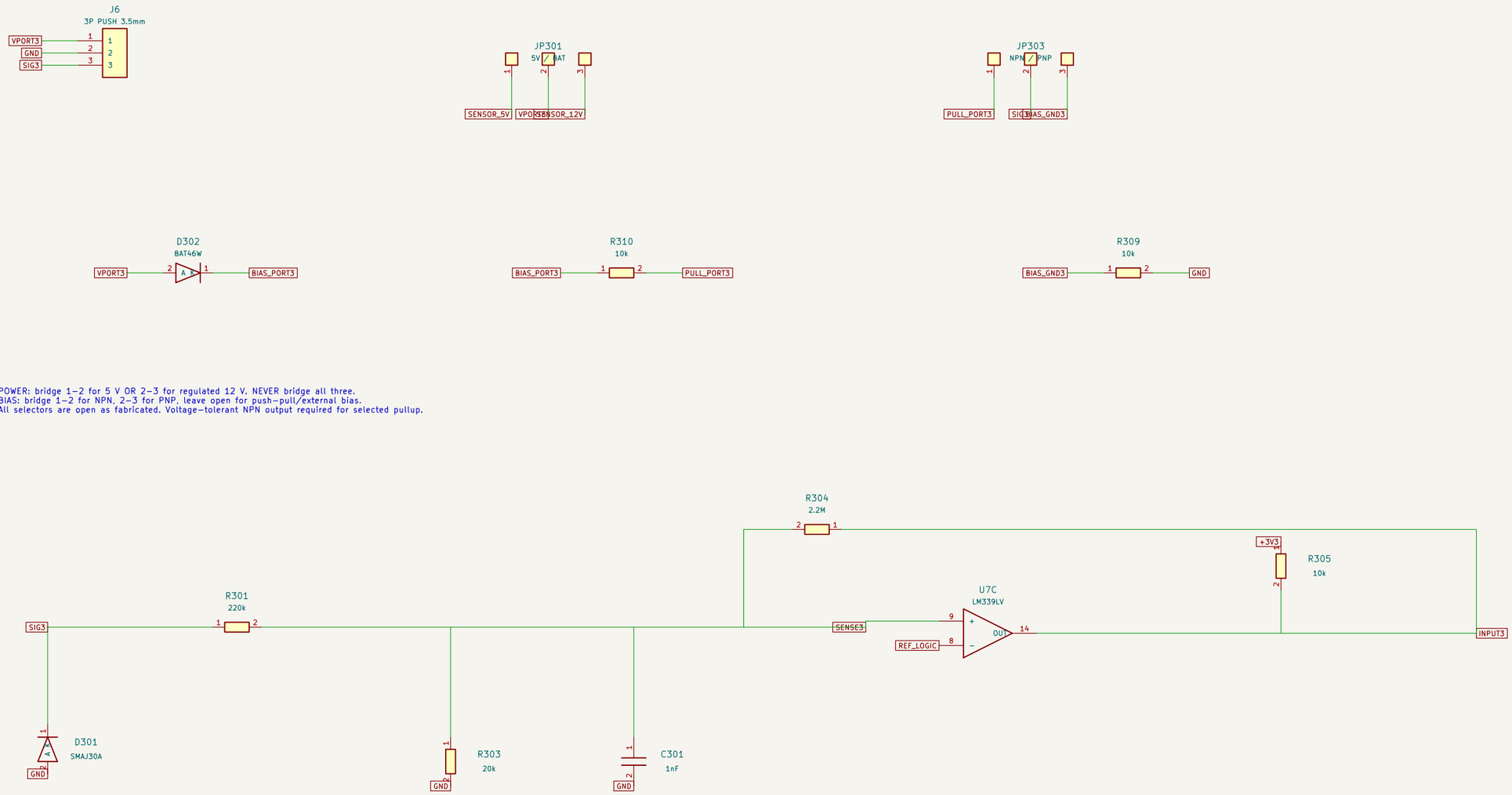
0–30 V signal design range; software debounce follows the hardware filter.
Nominal thresholds near 2.0 V rising / 1.65 V falling. Final bounds are in electrical_checks.json.
No analog measurement, 1.8 V logic guarantee or CTRE protocol compatibility.

Prototype: bench validation required before robot use
**Original sensor interface design**

| | | |
|---|---|---|
| Sheet: /Input 2: fixed-threshold digital input/ | | |
| File: input2.kicad_sch | | |
| ***Title: SC12 – Input 2: fixed-threshold digital input*** | | |
| Size: A3 | Date: 2026-09-08 | **Rev: D** |
| KiCad E.D.A. 10.0.6 | | Id: 8/9 |

# Input 3: fixed-threshold digital input

## SOLDER CONFIGURATION: SELECT POWER AND SENSOR OUTPUT TYPE




POWER: bridge 1-2 for 5 V OR 2-3 for regulated 12 V. NEVER bridge all three.
BIAS: bridge 1-2 for NPN, 2-3 for PNP, leave open for push-pull/external bias.
All selectors are open as fabricated. Voltage-tolerant NPN output required for selected pullup.

The reference divider/filter and comparator power connections are on input1.
One fixed logic-compatible threshold for all three channels; no industrial threshold selector.

0-30 V signal design range; software debounce follows the hardware filter.
Nominal thresholds near 2.0 V rising / 1.65 V falling. Final bounds are in electrical_checks.json.
No analog measurement, 1.8 V logic guarantee or CTRE protocol compatibility.

Prototype: bench validation required before robot use
**Original sensor interface design**

| | | |
|---|---|---|
| Sheet: /Input 3: fixed-threshold digital input/ | | |
| File: input3.kicad_sch | | |
| ***Title: SC12 - Input 3: fixed-threshold digital input*** | | |
| Size: A3 | Date: 2026-09-08 | **Rev: D** |
| KiCad E.D.A. 10.0.6 | | Id: 9/9 |